

DRP Fall 2020 Outline

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To-dos.

- Background: groups, fields, projective space.
- SAGE demonstrations: What kinds?
- Add diagrams.
- Fill in “ADD...”

Washington.

Chapter 2 (basics)

Weierstrass equation

In general, over field K , ADD...

For char $K \neq 2, 3$, a cubic reduces to Weierstrass form $E : y^2 = x^3 + Ax + B$ with discriminant $D = -(4A^3 + 27B^2) \neq 0$.

Group law

Let ∞ be identity. If $P_1 = (x_1, y_1)$, $P_2 = (x_2, y_2)$, then the sum $P_3 = (x_3, y_3) = P_1 + P_2$ has coordinates $x_3 = m^2 - x_1^2 - x_2^2$, $y_3 = \dots$, where m is slope of P_1P_2 . $(E(K), +)$ is an abelian group.

Group has different structure over different fields K . $K = \mathbb{F}_q$: finite abelian, $K = \mathbb{Q}$: finitely generated abelian, $K = \mathbb{C}$: a torus.

(Nowadays, no-one uses abelian groups for cryptography, due to quantum computing. Post-quantum cryptography will work with nonabelian groups.)

Endomorphisms

Definition. $\alpha : E(\overline{K}) \rightarrow E(\overline{K})$ is an endomorphism if it's a homomorphism whose coordinates are rational functions. (Thus, $\alpha = (R_1(x, y), R_2(x, y))$.)

You can show any endomorphism can be written as $\alpha(x, y) = (r_1(x), r_2(x)y)$.

Endomorphisms will be used to study subgroups of $E(K)$ (specifically, their kernels). Two important ones are the Frobenius map and the $\cdot n$ map.

Definition. Frobenius map $\phi_q : E(\overline{\mathbb{F}_q}) \rightarrow E(\overline{\mathbb{F}_q})$ is given by $\phi_q(x, y) = (x^q, y^q)$.

By finite field theory, ϕ_q only fixes $\mathbb{F}_q$. Hence $\ker(\phi_q - 1) = E(\mathbb{F}_q)$. To be generalized and used later, for Hasse.

Definition. $\cdot n$ map $\cdot n : E(\overline{K}) \rightarrow E(\overline{K})$ is given by $P \mapsto nP$.

Note $\ker(\cdot n) = E[n]$, points of order dividing n .

By studying when endomorphisms are separable, we get more information on the size of their kernel in relation to the degree... ADD (intuitive understanding? roots of denominator of $r_1(x)$?)

Singular ECs

ADD...

Chapter 3 (torsion points)

Structure of $E[n]$

Definition. $E[n]$ is the set of n torsion points, i.e. all P with $nP = 0$, i.e. all points of order dividing n .

Theorem (structure of $E[n]$). if $\text{char } K \nmid n$, then $E[n] \simeq \mathbb{Z}_n \oplus \mathbb{Z}_n$.

The proof is complicated. The key insights are: $E[n] = \ker(\cdot n)$, $\cdot n$ is a separable endomorphism of degree n^2 , thus $\#E[n] = n^2$.

Homomorphism as matrix

Note α maps $E[n]$ into $E[n]$ (as $nP = 0 \implies n\alpha(P) = 0$). Further α is linear on $E[n]$. Thus, once we choose a basis (say, $(1, 0), (0, 1)$) for $E[n] \simeq \mathbb{Z}_n \oplus \mathbb{Z}_n$, we can represent α as a matrix.

(In general, the Fundamental Theorem of Finitely Generated Abelian Groups says any group can be written as $G \simeq \mathbb{Z}^r \oplus G_{\text{torsion}}$, where $G_{\text{torsion}} \simeq \text{ADD} \dots$)

Chapter 4 (ECs over finite fields)

Structure of $E(\mathbb{F}_q)$

What's the structure of the finite abelian group $E(\mathbb{F}_q)$?

Theorem. $E(\mathbb{F}_q) \simeq \mathbb{Z}_n$ or $\mathbb{Z}_{n_1} \oplus \mathbb{Z}_{n_2}$.

Proof. ADD...

Size of $E(\mathbb{F}_q)$

Bounding group order

What is its size? We can provide a bound.

Hasse bound. $|a| \leq 2\sqrt{q}$, where $a = q + 1 - \#E(\mathbb{F}_q)$.

Thus $\#E(\mathbb{F}_q)$ lies in the Hasse interval, $[(\sqrt{q} - 1)^2, (\sqrt{q} + 1)^2]$.

The proof is complicated, but similar to showing $\#E[n] = n^2$. The key insights are: $E(\mathbb{F}_q) = \ker(\phi_q - 1)$, $\phi_q - 1$ is a separable endomorphism of degree $q + 1 - a$, thus $\#E(\mathbb{F}_q) = q + 1 - a$.

As a side result, we get $\phi_q^2 - a\phi_q + q = 0$ as endomorphisms. $x^2 - ax + q$ is the characteristic polynomial of Frobenius.

Determining group order

There are many methods for finding the exact size $\#E(\mathbb{F}_q)$.

The method of subfield curves: find $\#E(\mathbb{F}_{q^n})$ from $\#E(\mathbb{F}_q)$ recursively.

Theorem. If $x^2 - ax + q = (x - \alpha)(x - \beta)$, then $\#E(\mathbb{F}_{q^n}) = q^n + 1 - (\alpha^n + \beta^n)$.

Proof. Characteristic polynomial of Frobenius... ADD

Lang Trotter method: a simple explicit count with Legendre symbols.

Theorem. $\#E(\mathbb{F}_q) = q + 1 + \sum \left(\frac{x^3 + Ax^2 + B}{q} \right)$.

Using points of large order: if there's a point P of large order, since Lagrange's Theorem implies $\text{ord } P \mid \#E(\mathbb{F}_q)$, there are few possibilities for $\#E(\mathbb{F}_q)$ in the Hasse interval. Best case: only 1 multiple of $\text{ord } P$!

For $q = p^1$, this best case happens on either E or its twist E' (ADD...):

Mestre's theorem. For $p > 229$, either $E/\mathbb{F}_p$ or $E'/\mathbb{F}_p$ has a point $\text{ord } P$ with 1 multiple in the Hasse interval

Proof. Involves structure of $E(\mathbb{F}_p)$ and lots of bounding... ADD

Maybe we can't do that good for $\mathbb{F}_q$ in general. But we can still find points of large order.

Suppose $E(\mathbb{F}_q) \simeq \mathbb{Z}_{n_1} \oplus \mathbb{Z}_{n_2}$. Then lcm of two random points is likely n_2 , the maximum order. ADD...

Cremona–Harley theorem. n_2 determines n_1 ... ADD

(We've ignored the case when $E(\mathbb{F}_q) \simeq \mathbb{Z}_n$, as it's rare.)

Baby-step giant-step algorithm (BSGS). To find $\text{ord } P$.

The steps of the algorithm are as follows:

Find M :

- Let $Q = (q + 1)P$.
- Choose $m > q^{1/4}$ and compute $\pm jP$ for $0 \leq j \leq m$.
- Compute $Q + k(2mP)$ for $-m \leq k \leq m$, until there is a match with the list $\{\pm jP\}$.
- Note that $MP = 0$, where $M = q + 1 + 2mk \mp j$.

Find $\text{ord } P$:

- Let $M = p_1^{e_1} \cdots p_t^{e_t}$.
- Find $\frac{M}{p_i}P$ for all i ; if $\frac{M}{p_i}P = 0$ for some i , repeat these steps with $M \rightarrow \frac{M}{p_i}$. Else, $M = \text{ord } P$.

Supersingular curves (computational advantage). ADD...

Interlude, Chapter 3.3: Weil pairing

ADD...

Chapter 6 (ECC)

Cryptography

Setup. Use public key to setup symmetric key. Most protocols use the difficulty of the discrete log problem (DLP).

Diffie–Hellman encryption. A, B want to communicate. They have secret keys a, b , and a public key $P \in E/\mathbb{F}_q$. A sends aP to B , and B sends bP to A . Then both compute $Q = abP$ and extract a common key.

It's hard to find abP because that requires finding a or b , which is hard.

Massey–Omura encryption. Similar to the famous king communication puzzle (?). A wants to send B a message $P \in E(\mathbb{F}_q)$. A, B have secret keys $m_A, m_B \in \mathbb{Z}_N$, where $N = \#E(\mathbb{F}_q)$. He “locks” with m_A by sending m_AP to B . Then B “locks” with m_B and sends $m_B m_AP$ back to A . Then A “unlocks” by sending $m_BP = m_A^{-1}(m_B m_AP)$. Finally, B retrieves the original message by “unlocking” $P = m_B^{-1}(m_BP)$.

Boneh–Franklin encryption (identity-based). ... ADD

Digital signatures

Setup. How to sign a document.

ElGamal digital signature. A wants to sign a document, and B wants to verify.

Initialize:

- Choose $E/\mathbb{F}_q$, $A \in E(\mathbb{F}_q)$ with $\text{ord } A = N$, $B = aA$, $f : E(\mathbb{F}_q) \rightarrow \mathbb{Z}$ with few collisions
- Release all of these except the secret key a

A signs:

- Represents message as integer m with hash H
- Chooses a random k with $(k, N) = 1$ and computes $R = kA$
- Computes integer $s \equiv k^{-1}(m - af(R)) \pmod{N}$
- Releases signed message (m, R, s)

B verifies:

- Computes $V_1 = f(R)B + sR$
- Computes $V_2 = mA$
- Checks if $V_1 = V_2$

Important: different k 's must be chosen each time (?)

Digital Signature Algorithm (ECDSA). Variant of ElGamal that is quicker to verify.

Initialize:

- Choose $E/\mathbb{F}_q$ with $\#E(\mathbb{F}_q) = fr$ for f small and r prime, $G \in E(\mathbb{F}_q)$ with $\text{ord } G = r$, $Q = aG$
- Release all of these except the secret key a

A signs:

- Represents message as integer m with hash H
- Chooses a random $k \in [r]$ and computes $R = kG = (x, y)$
- Computes integer $s \equiv k^{-1}(m + ax) \pmod{r}$
- Releases signed message (m, R, s)

B verifies:

- Computes $u_1 \equiv s^{-1}m \pmod{r}$
- Computes $u_2 \equiv s^{-1}x \pmod{r}$
- Checks if $R = u_1G + u_2Q$

Exposition from Katz Lindell

Definition 0.1 (3-round identification scheme).

- The prover generates an initial message $I = P_1(sk)$ and sends it to the verifier.
- The verifier generates a challenge r based on pk and sends it to the prover.
- The prover generates a response $s = P_2(sk, r)$ and sends it to the verifier.
- The verifier checks that $V(pk, r, s) = I$; if this is true, the prover is authenticated.

Definition 0.2 (Fiat-Shamir transform).

- Generate an initial message I .
- Generate a challenge $r = H(I, m)$, where H is a hash function.
- Generate a response s to the challenge.
- Sign your message with (r, s) .

To verify the signature, a person computes $I = V(pk, r, s)$ and verifies that $H(I, m) = r$.

Definition 0.3 (Schnorr identification scheme).

- The prover generates an initial message $I = g^k$ for a random $k \in \mathbb{Z}_p^\times$ and sends it to the verifier.
- The verifier generates a challenge $r \in \mathbb{Z}_p^\times$ at random and sends it to the prover.
- The prover generates a response $s \equiv rx + k \pmod{p}$ and sends it to the verifier.
- The verifier checks that $g^s \cdot y^{-r} = I$; if this is true, the prover is authenticated.

Definition 0.4 (DSA/ECDSA identification scheme).

- The prover generates an initial message $I = g^k$ for a random $k \in \mathbb{Z}_p^\times$ and sends it to the verifier.
- The verifier generates a challenge $r, \alpha \in \mathbb{Z}_p$ at random and sends it to the prover.
- The prover generates a response $s \equiv k^{-1}(\alpha + rx) \pmod{p}$ and sends it to the verifier.
- The verifier checks that $s \neq 0$ and $g^{\alpha s^{-1}} \cdot y^{rs^{-1}} = I$; if this is true, the prover is authenticated.

Fiat–Shamir variant.

- Generate an initial message I .
- Generate a challenge $r = F(I)$ and $\alpha = H(m)$, where $F : G \rightarrow \mathbb{Z}_p$ and H are hash functions.
- Generate a response s to the challenge.
- Sign your message with (r, s) .

Chapter 5 (attacks on DLP)

General attacks

Index calculus. For $\mathbb{F}_p^\times$, given a generator g , how to compute $L(h) = \log_g h$?

Idea: factor $h = p_1^{e_1} \cdots p_k^{e_k}$ and find $L(p_i)$ instead!

- Choose factor base $B = \{\text{small primes } p_i\}$
- Compute $g^0, g^1, \dots$ to find a system of equations in $L(p_i)$
- Solve for $L(p_i)$ with linear algebra
- Compute $hg^0, hg^1, \dots$ until it factors into primes in B and solve for $L(h)$

This can be extended to $\mathbb{F}_q^\times$ with algebraic NT.

For the next few sections, we consider the general DLP: given group G with $\#G = N$, P , and Q , find k .

BSGS. Last time, used BSGS to find $\text{ord } P$, i.e. M such that $MP = \infty$. Now, we modify it to find k such that $kP = Q$.

- Fix $m \geq \sqrt{N}$, find mP
- Store iP for $0 \leq i < m$ (baby step)
- Compute $Q - jmP$ for $0 \leq j < m$ until it matches something in $\{iP\}$ (giant step)
- Return $k \equiv i + jm \pmod{N}$

This works because $k = m(\text{quotient}) + \text{remainder}$, where $\text{quotient}, \text{remainder} < m$.
 $O(\sqrt{N})$ time/space.

Pollard ρ . This offers a space improvement over BSGS.

Take a random walk around G . Choose random function $f : G \rightarrow G$. Starting from P_0 , generate sequence $P_{i+1} = f(P_i)$. The sequence is periodic, hence $P_i = P_j$ for some i, j . (That's why it's the ρ method.)

How do we choose f to solve DLP?

- Partition $G = S_1 \sqcup S_2 \cdots \sqcup S_s$
- Choose $2s$ random $a_i, b_i \pmod{N}$
- Let $M_i = a_i P + b_i Q$ (possible steps)
- Define f piecewise by $f(g) = g + M_i$ if $g \in S_i$

When we find $P_i = P_j$, we know $c_i P + d_i Q = c_j P + d_j Q \implies mQ = nP$ for some m, n . Only finitely many possible k .

Pollard ρ can be optimized by storing fewer points.

Pohlig–Hellman. A base q approach (?).

Idea: suppose $\text{ord } P = N = q_1^{e_1} \cdots q_k^{e_k}$. Let's find $k \pmod{q_i^{e_i}}$ for all i , then use CRT to determine $k \pmod{N}$, i.e. k .

For a given $q^e = q_i^{e_i}$, imagine $k = k_0 + k_1 q + k_2 q^2 + \dots$ is its base- q representation:

- Compute $T = \{k(\frac{N}{q}P) \mid 0 \leq k < q\}$.

Let $Q_0 = Q$; for $1 \leq r < e$:

- Compute $Q_r = Q_{r-1} - k_{r-1} q^{r-1} P$.
- Find $k_r(\frac{N}{q}P) \in T$ such that $\frac{N}{q^{r+1}} Q_r = k_r(\frac{N}{q}P)$.

Once done, return $k_0 + k_1 q + \dots \pmod{q^e}$.

Fix order of iteration... ADD

EC specific attacks

Decision Diffie–Hellman attack. Although DH is hard, a modified version of DH can be solved: decision DH.

Decision DH: given P, Q , decide if $Q = abP$.

This can be solved by defining the modified Weil pairing $\tilde{e}_n \dots$ ADD

MOV attack. Reduces DLP in $E(\mathbb{F}_q)$ to DLP in $\mathbb{F}_{q^m}^\times$.

Setup: given $P, Q \in E(\mathbb{F}_q)$, $\text{ord } P = N$, $(q, N) = 1$, find k such that $Q = kP$.

Know that e_n tells us when k exists.

- Choose m such that $E[N] \subset E(\mathbb{F}_{q^m})$. Then $\mu_N \subset E(\mathbb{F}_{q^m})$

We first find a random point $T_1 \in E[N]$.

- Choose random $T \in E(\mathbb{F}_{q^m})$
- Find $M = \text{ord } T$
- Let $d = (M, N)$ and compute $T_1 = \frac{M}{d} T$. Then $\text{ord } T_1 = d \mid N \implies T_1 \in E[N]$

Now we find $k \pmod{d}$.

- Find $\zeta_1 = e_N(P, T_1)$ and $\zeta_2 = e_N(Q, T_1)$; these are both in $\mu_d \subset E(\mathbb{F}_{q^m})$
- Solve the DLP $\zeta_2 = \zeta_1^k$ in $\mathbb{F}_{q^m}^\times$ to find $k \pmod{d}$
- Combine $k \pmod{d}$ for many d to get $k \pmod{N}$, i.e. k

To avoid having to work in $\mathbb{F}_{q^m}$ with m large, use supersingular curves, which guarantee $m = 2$.

Supersingular curves are good for encryption, but also good for attacks... ADD

Anomalous-curve attack. ADD...

Chapter 7 (other applications)

Interlude, Chapter 2.11: ECs mod n

ADD... (overlap with Silverman?)

Factoring

$p - 1$ method, Lenstra's algorithm... ADD

Primality testing

Pohlig Lehrman test, EC version?... ADD

Chapter 11 (divisors)

Goal of chapter: how to construct/compute the Weil pairing.

Divisors

First, we introduce divisors.

Definition. A divisor D is a formal linear combination $a_1[P_1] + \dots + a_m[P_m]$ for $P_i \in E(\overline{K})$.

Definition. The degree of a divisor is $\deg D = \sum a_j$.

Definition. The sum of a divisor is $\text{sum}(D) = \sum a_j P_j$.

Definition. The group $\text{Div}^0 E$ is the set of degree 0 divisors.

Functions

A seemingly unrelated concept is that of a function on an EC.

Definition. A function $f : E(\overline{K}) \rightarrow \overline{K}$ is a rational function $f(x, y) = \frac{a(x, y)}{b(x, y)}$.
We often homogenize f .

Definition. $P = (x : y : z)$ is a zero of f if $a(x : y : z) = 0$.

Definition. $P = (x : y : z)$ is a pole of f if $b(x : y : z) = 0$.

Definition. The order $\text{ord}_P(f)$ of f at P is the multiplicity of P in the numerator/denominator, or 0 otherwise.

Principal divisors

Now comes the connection between divisors and rational functions.

Definition. The divisor of f is $\text{div}(f) = \sum_{P \in E(\overline{K})} \text{ord}_P(f) \cdot [P]$. Such a divisor is called a principal divisor.

Properties of principal divisors:

- $\text{div}(f)g = \text{div}(f) + \text{div}(g)$
- $\text{div}(f) = 0 \iff f \equiv c$ (complex analysis, but intuitive)
- $\text{div}(f) = \text{div}(g) \iff f = cg$
- $\text{div}(f) \in \text{Div}^0 E$ for all f (complex analysis, but kinda intuitive)

It turns out that there is an easy classification for principal divisors.

Principal Divisor Theorem. suppose $D \in \text{Div}^0 E$. Then D principal iff $\text{sum}(D) = \infty$.

Proof. Key idea: express $[P_1] + [P_2]$ in terms of $[P_1 + P_2]$ (“addition formula”). There are two functions (both lines) involved.

Consider line containing $P_1, P_2, P_3 = -(P_1 + P_2)$ with equation $ax + by + c = 0$. Then

$$\text{div}(ax + by + c) = [P_1] + [P_2] + [P_3] - 3[\infty].$$

Also, using the vertical through $P_3 = (x_3, y_3)$,

$$\text{div}(x - x_3) = [P_3] + [-P_3] - 2[\infty].$$

Subtracting, we get

$$[P_1] + [P_2] = [P_1 + P_2] + [\infty] + \text{div}\left(\frac{ax + by + c}{x - x_3}\right).$$

(A trick we used: figure out order of ∞ by requiring $\text{sum} = 0$.)

From here, the proof follows from repeatedly applying the addition formula, bunching up the $a_j[P_j]$ ’s in D with $a_j > 0$ and similarly for $a_j < 0$.

Note that the proof provides a way to find f such that $D = \text{div}(f)$. Unfortunately, it’s slow. (But it generalizes to curves besides ECs.)

Weil pairing

Back to the main point – defining the Weil pairing.

Suppose $T \in E[n]$. By the Principal Divisor Theorem, there exists f such that $\text{div}(f) = n[T] - n[\infty]$.

Let $T' \in E[n^2]$ such that $T = nT'$. Then by the Principal Divisor Theorem, there exists g such that

$$\text{div}(g) = \sum_{R \in E[n]} [T' + R] - \sum_{R \in E[n]} [R].$$

This is independent of T' (?).

Define the composition $f \circ n$, where n is $\cdot n$. Then $\text{div}(f \circ n) = \text{div}(g^n) \implies f \circ n = g^n$, up to scaling.

Finally, note that if $P \in E(\overline{K})$, $S \in E[n]$, we may define $\frac{g(P+S)}{g(P)} \in \mu_n$. This is independent of P by analysis (?).

Definition. The Weil pairing is $e_n(S, T) = \frac{g(P+S)}{g(P)}$.

Properties: ADD...

Note that computing the Weil pairing requires us to find g such that $\text{div}(g) = \text{sum of } n^2 \text{ things}$. Our approach from the proof of the Principal Divisor Theorem is $O(N^2)$, too slow!

There's a slightly more efficient definition of the Weil pairing.

Definition. Let $S, T \in E[n]$. Let $D_S, D_T \in \text{Div}^0 E$. Suppose f_S, f_T are such that $\text{div}(f)_S = nD_S$ and $\text{div}(f)_T = nD_T$. Then $e_n(S, T) = \frac{f_T(D_S)}{f_S(D_T)}$. (?)

A natural choice of D_S, D_T is $D_S = [S] - [\infty]$ and $D_T = [T + R] - [R]$.

Miller's algorithm

Now we only need an algorithm that can solve $\text{div}(f) = n[P + R] - n[R]$ fast, and evaluate $f(Q_1)/f(Q_2)$ as well.

Miller's algorithm. Initial idea: compute f_j such that $\text{div}(f)_j = j[P + R] - j[R]$, and use successive doubling to get to $f_n = f$. (In fact, don't even compute f_j explicitly; just the fraction needed!)

But $\text{sum } j[P + R] - j[R] \neq 0!$ Add terms: $\text{div}(f)_j = j[P + R] - j[R] - [jP] - [\infty] = D_j$.

Key formula: given $f_j(Q_1)/f_j(Q_2)$ and $f_k(Q_1)/f_k(Q_2)$, can quickly find $f_{j+k}(Q_1)/f_{j+k}(Q_2)$.

Why? From the addition formula,

$$\begin{aligned} D_{j+k} &= D_j + D_k + \text{div}(h) \implies \text{div}(f)_{j+k} = \text{div}(f)_j f_k h \implies f_{j+k} = f_j f_k h \\ &\implies \frac{f_{j+k}(Q_1)}{f_{j+k}(Q_2)} = \frac{f_j(Q_1) f_k(Q_1) \text{stuff}}{f_j(Q_2) f_k(Q_2) \text{stuff}} \end{aligned}$$

Chapters 9 & 10 (CM: complex multiplication)

ECs over $\mathbb{C}$

Goal. Show that an elliptic curve is the same as a complex torus.

First, we show any torus yields an elliptic curve.

It turns out we can parametrize complex points on an EC with the ρ function.

Definition. Given a lattice $L = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$, the Weierstrass ρ function is

$$\rho(z) = \frac{1}{z^2} + \sum_{\omega \in L, \omega \neq 0} \left(\frac{1}{(z - \omega)^2} - \frac{1}{\omega^2} \right).$$

ρ is doubly periodic with periods ω_1, ω_2 ; hence it induces a map from $\mathbb{C}/L \rightarrow \mathbb{C}$.

We can expand the reciprocals in ρ to get a Laurent series. The coefficients are in terms of the Eisenstein series $G_k = \sum_{\omega \in L, \omega \neq 0} \frac{1}{\omega^k}$.

Lemma. $\rho(z) = \frac{1}{z^2} + \sum (2j+1)G_{2j+2}z^{2j}$.

With the above lemma, the following differential equation follows.

Lemma. $\rho'^2 = 4\rho^3 - g_2\rho - g_3$, where $g_2 = 60G_4$ and $g_3 = 140G_6$.

It can be shown that $\Delta \neq 0$ for the cubic. But this is the equation of an EC!

Theorem. The map $\Phi : \mathbb{C}/L \rightarrow E(\mathbb{C})$ given by $\Phi : z \mapsto (\rho(z), \rho'(z))$ is an isomorphism.

Proof. Mainly need to check Φ is a homomorphism... ADD

Next, we show that any elliptic curve comes from a torus.

Theorem. Given the EC $y^2 = 4x^3 - Ax - B$, there exists a lattice L such that $g_2(L) = A$ and $g_3(L) = B$. Further, there is an isomorphism $\mathbb{C}/L \simeq E(\mathbb{C})$.

Idea: given an EC $y^2 = 4x^3 - Ax - B$, find L such that $g_2(L) = A$ and $g_3(L) = B$.

It turns out we can do this if we just match the j invariant of E with the j invariant of (a scaled version of) L .

But what is $j(L)$? It's what you'd expect. Definition: $j(L) = 1728 \frac{g_2^3}{\Delta}$.

To find an expression for $j(L)$, we can find expansions for $g_2(L)$ and $g_3(L)$. To do this, it's convenient to rewrite as follows. Let $L_\tau = \mathbb{Z}\tau + \mathbb{Z}$, where WLOG $\tau = \omega_1/\omega_2 \in \mathcal{H}$. Then $g_i(\tau) = g_i(L_\tau)$.

Lemma. $g_2(\tau) = \frac{4\pi^4}{3}(1 + 240q + \dots)$ and $g_3(\tau) = \frac{8\pi^6}{27}(1 + 504q + \dots)$, where $q = e^{2\pi i\tau}$.

Proof. Differentiate the Euler product formula for $\sin$ to find an expression for G_{2k} .

Lemma. $j(\tau) = j(L) = \frac{1}{q} + 744 + 196884q + \dots$

Proof. Divide the two expansions. To see why j is unchanged under homothety, note that g_2 and g_3 's scalings cancel out in its expression.

j satisfies some other properties related to the modular group $\mathrm{SL}_2(\mathbb{Z})$. [Other interesting properties?... ADD]

Lemma. $j(M\tau) = j(\tau)$ for all $M = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in \mathrm{SL}_2(\mathbb{Z})$, where $M\tau = \frac{a\tau+b}{c\tau+d}$ is a Mobius transformation.

Proof. See how G_{2k} changes.

It turns out j is meromorphic with fundamental domain $\mathcal{F}$... ADD? By restricting to $\mathcal{F}$, we also get the following.

Theorem. $\mathrm{ord}_{i\infty} f + \frac{1}{3} \mathrm{ord}_\rho f + \frac{1}{2} \mathrm{ord}_i f + \sum \mathrm{ord}_z f = 0$, where $\rho = e^{2\pi i/3}$.

Proof. Follows more generally from functions defined on Riemann surfaces? Directly, use complex integration... ADD?

As a result, we get find that $j : \mathcal{F} \rightarrow \mathbb{C}$ is bijective.

Theorem. For all $z \in \mathcal{H}$, there exists a unique $\tau \in \mathcal{F}$ such that $j(\tau) = z$.

Theorem. Let $\tau_1, \tau_2 \in \mathcal{H}$. Then $j(\tau_1) = j(\tau_2)$ iff there exists $M \in \mathrm{SL}_2(\mathbb{Z})$ such that $M\tau_1 = \tau_2$. We can even translate this back in terms of lattices.

Theorem. Let L_1, L_2 be lattices. Then $j(L_1) = j(L_2)$ iff L_1 is homothetic to L_2 .

Finally, we're ready to prove that every EC comes from a lattice!

Proof. Let $j = 1728 \frac{A^3}{\Delta}$ be the j invariant of E . By our result, we may choose τ such that $j(\tau) = j(L_\tau) = j$. Next, if $g_2(L) \neq 0$, choose λ such that $g_2(\lambda L) = A$. Then, either λL works or $i\lambda L$ works. The case $g_2(L) = 0$ is similar.

We can now think geometrically about n torsion points of E . These are $\{\frac{j}{n}\omega_1 + \frac{k}{n}\omega_2\}$.

CM for ECs over $\mathbb{C}$

CM occurs when the endomorphism ring $\text{End}(E)$ is larger than $\mathbb{Z}$. We know that $\cdot n \in \text{End}(E)$ for all n , hence $\mathbb{Z} \subset \text{End}(E)$. But how much bigger is it?

For E over $\mathbb{C}$, if $\alpha \in \text{End}(E)$, we get a corresponding homomorphism $\tilde{\alpha} : \mathbb{C}/L \rightarrow \mathbb{C}/L$. This is analytic on $U \ni 0$ (?) and satisfies Cauchy's functional equation, hence $\tilde{\alpha}(z) = \beta z$ for some $\beta \in \mathbb{C}$. The following tells us that these are all the endomorphisms.

Theorem. If E is an EC over $\mathbb{C}$ with lattice L , then $\text{End}(E) \simeq \{\beta \in \mathbb{C} \mid \beta L \subset L\}$.

We can even say more about $\text{End}(E)$. It turns out $\text{End}(E)$ is either $\mathbb{Z}$ or an order in an imaginary quadratic field. What's the latter? An imaginary quadratic field is simply $K = \mathbb{Q}(\sqrt{-D})$. The ring of integers O_K is $\mathbb{Z}[\delta]$ for $\delta = \dots$ ADD. It's all the algebraic integers in K . Finally, an order is simply a subring $\mathbb{Z} \subset R \subset O_K$.

Theorem. $\text{End}(E)$ is either $\mathbb{Z}$ or an order in an imaginary quadratic field.

Proof. ADD...

CM for ECs over finite fields

ADD...

$j(L)$ is an algebraic integer

ADD...

Silverman:

ECs over $\mathbb{Q}$

Nagell Lutz, Mordell Weil, Mazur... ADD

Mordell Weil Theorem

Mordell–Weil Theorem. $E(\mathbb{Q})$ is finitely generated.

Key idea: given $P \in C(\mathbb{Q})$, use descent to write as sum of elements of a generating set.

Need measure of points. Use the height:

- Height of rational: $H(m/n) = \max(|m|, |n|)$
- Height of point: $H(x, y) = H(x)$
- “Small H ” height: $h = \log H$

Need 4 lemmas.

Lemma 0.5 (Lemma 1, finiteness property)

$\forall M, \{x \in C(\mathbb{Q}) \mid h(x) \leq M\}$ finite

Proof. Clear, from finiteness property of $h(m/n)$. □

Lemma 0.6 (Lemma 2)

$\forall P_0, \exists k_0$ depending on P_0 s.t. $\forall P, h(P + P_0) \leq 2h(P) + k_0$

Lemma 0.7 (Lemma 3)

$\exists k$ s.t. $\forall P, h(2P) \geq 4h(P) - k$

Lemma 0.8 (Lemma 4)

$[C(\mathbb{Q}) : 2C(\mathbb{Q})]$ finite.

We may rephrase Mordell Weil more generally.

Theorem 0.9

Γ abelian group. $h : \Gamma \rightarrow [0, \infty)$ function. Suppose Lemmas 1-4 hold. Then Γ is finitely generated.

Proof. Suppose $P \in \Gamma$. We apply descent to reduce P to sum of elements of generating set.

Choose representatives $Q_1, \dots, Q_n$ of cosets of 2Γ . Suppose $P \in Q_{i_1} + 2\Gamma$, i.e. $P = Q_{i_1} + 2P_1$ for some i_1, P_1 . Repeat on P_1 , then on P_2 , etc. to find:

$$P_1 = Q_{i_2} + 2P_2$$

$$P_2 = Q_{i_3} + 2P_3$$

...

$$P_{m-1} = Q_{i_m} + 2P_m$$

Substituting, find $P = \sum 2^{j-1} Q_{i_j} + 2^m P_m$.

Claim: $\forall P$, there exists m s.t. $h(P_m) < \epsilon$ for a fixed ϵ . (Hence, we can reduce to P_m in finite set.)

To see this, use Lemmas 1-3:

Know $\exists k'$ s.t. $h(P - Q_i) \leq 2h(P) + k' \forall P, i$

Hence know $4h(P_j) \leq h(2P_j) + k = h(P_{j-1} - Q_{i_j}) + k \leq 2h(P_{j-1}) + k' + k$ (i.e. $h(P_j) \approx \frac{1}{2}h(P_{j-1})$)

There exists m s.t. $h(P_m) \leq k + k'$, as desired.

Follows that can reduce P to an element of finite gen set $\{Q_i\} \cup \{P_m\}$, as desired. $\square$

Height of $P + P_0$

ADD...

Background**Group theory**

Definition of a group

Abelian and non abelian groups

Cyclic groups

Subgroups

Homomorphisms and isomorphisms

Left and right cosets (affine subgroups):

Given $H \leq G$, the left cosets of H partition G .

Motivation: want to form group out of cosets. Ex: $H = 3\mathbb{Z}_{12} \leq G = \mathbb{Z}_{12}$.

Normal subgroup: $aH = Ha$ for all $a \in G$. (a 's left and right cosets are the same.)

- All subgroups of an abelian group are normal.

Quotient group: G/H , consisting of cosets aH with operation $(aH)(bH) = abH$.

Define $(\text{mod } H)$ equivalence relation. (Equal if differ by $h \in H$.) Equivalence classes are left cosets.

- Well defined? Yes...
- Work with representatives.
- Note: $\mathbb{Z}_{mn}/\mathbb{Z}_m \simeq \mathbb{Z}_n$.

Simple group: no normal subgroups other than $\{e\}$ and G . "Indivisible."

Quotient mapping theorem: Let $H \leq G$. Then H is normal **iff** there exists a homomorphism $\phi : G \rightarrow G/H$ with $\ker \phi = H$.

- $\{\text{normal subgroups}\} \iff \{\text{homomorphisms out of } G\}$
- $(\rightarrow)$ Naturally, $\phi(g) = gH$. $(\leftarrow)$...

First isomorphism theorem: Let $\phi : G \rightarrow H$ be a homomorphism. Then:

- $K = \ker \phi$ is a normal subgroup of G (see above).
- $I = \text{im } \phi$ is a subgroup of H (check).
- $I \simeq G/K$.
- Third point: quick way to characterize quotient groups. Ex...

Direct product/sum group: $G \otimes H = \{(g, h) \mid g \in G, h \in H\}$. Use $\oplus$ if abelian.

Finite groups: Lagrange's theorem

Sylow's theorem (partial converse) (?)

Structure theorem for finite abelian groups (see below)

Structure theorem for finitely generated abelian groups:

$$G \simeq \mathbb{Z}^r \oplus \mathbb{Z}_{p_1^{e_1}} \oplus \mathbb{Z}_{p_2^{e_2}} \cdots \oplus \mathbb{Z}_{p_k^{e_k}} \simeq \mathbb{Z}^r \oplus \mathbb{Z}_{n_1} \oplus \mathbb{Z}_{n_2} \cdots \oplus \mathbb{Z}_{n_k}$$

with $n_i \mid n_{i+1}$ for all i .

Examples of groups...

PROOFS?

Ring theory

Definition of a ring (commutative with 1)

Subring

Units: $U(R)$ is the multiplicative subgroup of elements with a multiplicative inverse.

Zero divisors: x is zero divisor if there exists $y \neq 0$ with $xy = 0$.

Group ring: given ring R and finite group G , define $RG \dots$

Integral domain (ID): a ring with no zero divisors. ("Field without inverses.")

- Can do cancellation.
- Finite ID = field.

Ideals:

Motivation: corresponding concept to normal subgroups. Invariant under conjugation $\implies$ invariant under multiplication.

Definition. Subring I is ideal if $rI \subset I$ for all $r \in R$.

- Ex: $n\mathbb{Z}$ in $\mathbb{Z}$.
- Note: if I contains a unit, $I = R$.

Quotient ring: can quotient by any subring R' to get additive quotient group R/R' . But when quotienting by ideal, get a quotient *ring* R/I .

Define (mod I) equivalence relation. (Equal if differ by $x \in I$.) Equivalence classes are translates rI .

Exs:

- $\mathbb{Z}/n\mathbb{Z}$, the ring $\mathbb{Z}_n$.
- $\mathbb{Z}_2[x]/(x^2 + x + 1) \simeq \{0 + I, 1 + I, x + I, (x + 1) + I\} = \{0, 1, x, x + 1\}$, a finite field with 4 elements.

Ring homomorphisms and isomorphisms: much carries over from groups.

Homomorphism: preserves $+$ and $\cdot$.

Exs:

- Reduction (mod n) map.
- Evaluation map by a polynomial $p(x) \in \mathbb{R}[x]$.

First isomorphism theorem: Let $\phi : R \rightarrow S$ be a homomorphism. Then:

- $K = \ker \phi$ is an ideal of R .
- $I = \text{im } \phi$ is a subring of S .
- $I \simeq R/K$.

Maximal ideal: I is maximal if the only bigger ideal containing it is R .

I is maximal iff R/I is a field.

Ex: $R = \mathbb{Z}_2[x]$, $I = (x^2 + x + 1)$; any irreducible?

[Prime ideal: P is prime if $ab \in P \iff a \in P$ or $b \in P$.

P is prime iff R/P is an ID. (Hence, maximal $\implies$ prime.) (?)]

Focusing on IDs:

We are used to $\mathbb{Z}, \mathbb{R}, \mathbb{C}$, so to study extensions, we focus on IDs. Many familiar properties fail in general, like unique factorization, Euclidean algorithm, etc.

Divisibility

a is an associate of b if $a \mid b$ and $b \mid a$.

Irreducible: only divisors are units and associates.

Primes: $p \mid ab \implies p \mid a$ or $p \mid b$.

In general, prime $\implies$ irreducible, but irreducible $\not\implies$ prime. (Ex: $\mathbb{Z}[\sqrt{-5}]$.)

For “nice” rings, principal ideal domains (PID), it does hold.

Definition. A principal ideal is of the form (a) .

Definition. A PID is an ID R whose ideals are all principal.

PIDs have gcds, lcms, and irreducible $\implies$ prime.

Exs: $\mathbb{Z}, F, F[x]$ (?). Not $\mathbb{Z}[x]$, since $(2, x)$ is not principal.

Lemma. If $a, b \in R^*$, then $\gcd(a, b)$ exists. Further, there exist x, y such that $ax + by = \gcd(a, b)$.

Corollary. Irreducible $\implies$ prime.

Another notion:

Definition. A UFD is an ID with unique factorization into irreducibles, i.e.:

(1) any element can be factored into a product of irreducibles.

(2) irreducible $\implies$ prime.

Exs: $\mathbb{Z}$ (FTA), $F[x, y], \mathbb{Z}[x]$.

Theorem. PID $\implies$ UFD. (?)

Another notion:

Definition. A Euclidean domain is an ID where there is an analogue of the Euclidean algorithm. (Need a Euclidean function to measure size in the division algorithm.) (purpose?)

Exs: $\mathbb{Z}, F[x], \mathbb{Z}[i]$.

Theorem. Euclidean $\implies$ PID.

Algebraic integers:

Definition. An algebraic integer is a root of a monic polynomial in $\mathbb{Z}[x]$.

Definition. The ring of integers O_K in the field $K = \mathbb{Q}(\sqrt{d})$ is the set of algebraic integers in K .

Properties of O_K ...

Norm in $\mathbb{Q}(\sqrt{d})$... when is O_K norm Euclidean?... when is O_K Euclidean?...

PROOFS?

Examples of rings...

Polynomial rings (see Gunning / Conrad)...

ADD Katy's notes from meetings...

Field theory

...

Examples of fields...

<https://www.math.utah.edu/~schwede/math435/index.html>